

GOLCUK DUMLUPINAR, TurkeyWMO: **170670**Lat: **40.7267N**Lon: **29.8067E**Elev: **18**StdP: **101.11**Time Zone: **3.00 (E03)**Period: **94-16**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	0.1	-4.3	2.6	0.9	-2.9	3.0	2.4	8.0	11.6	7.1	11.0	1.8	140	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	34.1	23.3	32.6	22.8	31.2	22.4	25.1	31.5	24.2	30.3	23.5	29.2	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	17.9	28.9	22.3	17.0	28.0	21.5	16.2	27.2	76.4	31.6	73.1	30.3	70.0	29.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.4	4.6	-3.1	37.8	2.1	1.8	-4.6	39.1	-5.8	40.2	-7.0	41.2	-8.5	42.5		
			-3.8	26.1	2.0	1.1	-5.3	26.9	-6.4	27.5	-7.5	28.1	-9.0	28.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	6.8	7.6	9.5	13.8	18.7	23.0	25.6	25.9	21.9	17.0	12.1	8.6
	DBStd	7.57	3.84	4.07	3.73	3.70	3.17	2.67	2.16	2.49	3.02	3.46	3.69	3.79
	HDD10.0	357	114	87	55	8	0	0	0	0	0	1	19	72
	HDD18.3	1669	357	299	273	142	34	1	0	0	6	66	188	302
	CDD10.0	2526	16	22	41	123	269	389	484	493	358	218	83	29
	CDD18.3	796	0	1	1	6	46	141	226	234	114	24	2	1
	CDH23.3	6432	1	2	4	49	308	1094	2019	2109	722	106	14	3
CDH26.7	2410	0	0	1	11	73	382	826	886	214	18	1	0	
Wind	WSAvg	1.9	1.8	2.0	2.2	2.1	1.9	1.8	1.9	1.9	1.7	1.6	1.6	1.8
Precipitation	PrecAvg	725	81	65	69	57	45	49	30	35	54	81	70	92
	PrecMax	1039	167	137	113	132	152	124	89	176	113	190	144	227
	PrecMin	501	25	11	23	7	5	2	1	0	1	5	7	36
	PrecStd	122	38	33	24	30	33	30	28	43	35	46	38	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	21.4	23.3	28.0	31.5	34.5	35.8	36.3	32.9	28.6	25.6	22.6
		MCWB	11.8	12.7	13.4	17.7	20.2	22.8	23.2	24.3	21.8	19.7	15.5	13.5
	2%	DB	16.9	18.3	19.7	24.3	28.6	32.2	33.6	34.2	30.7	26.0	21.7	18.5
		MCWB	10.7	11.7	12.4	15.9	19.3	22.1	23.1	23.7	21.4	18.4	15.0	12.4
	5%	DB	14.9	15.8	17.6	21.9	26.5	30.5	32.2	32.7	29.1	24.0	19.3	16.4
		MCWB	10.0	10.7	11.6	14.9	18.7	21.5	22.9	23.3	20.9	17.9	14.2	11.6
	10%	DB	12.5	13.6	15.6	20.0	24.8	28.9	30.9	31.2	27.5	22.1	17.4	14.2
		MCWB	9.2	9.8	10.7	14.1	18.1	20.9	22.5	22.7	20.3	17.5	13.6	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	13.9	14.1	18.4	21.5	24.7	25.9	26.2	24.1	21.1	17.2	14.5
		MCDB	18.0	19.2	19.8	26.1	29.0	31.9	33.1	33.2	30.2	26.0	21.7	19.6
	2%	WB	11.3	12.3	13.0	16.7	20.2	23.2	24.8	25.3	22.7	19.7	15.9	13.1
		MCDB	15.7	16.5	18.7	23.1	26.7	29.8	31.3	31.8	28.0	23.9	19.8	16.9
	5%	WB	10.3	11.1	12.0	15.5	19.3	22.4	24.0	24.5	21.8	18.6	14.9	12.2
		MCDB	13.9	15.2	16.6	21.0	25.4	28.6	30.3	30.5	26.9	22.5	18.2	15.5
	10%	WB	9.3	10.0	11.1	14.4	18.4	21.6	23.3	23.7	21.0	17.8	13.9	11.0
		MCDB	12.2	13.3	15.0	19.4	23.8	27.5	29.1	29.2	26.0	21.2	16.7	13.7
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.9	7.9	9.2	9.8	10.3	10.3	10.3	9.8	7.7	7.2	6.2
		MCDBR	9.7	10.6	11.9	12.9	13.1	13.1	12.4	12.4	12.2	10.9	10.2	8.9
		MCWBR	5.0	5.4	5.9	6.4	6.3	5.8	5.1	4.7	5.1	5.0	5.1	4.5
	5% WB	MCDBR	9.0	9.8	10.7	12.0	11.7	11.4	10.6	10.9	10.3	9.1	8.7	7.6
		MCWBR	5.0	5.3	5.4	6.3	6.0	5.6	5.0	4.7	4.7	4.6	4.7	4.2
Clear-Sky Solar Irradiance	taub	0.361	0.390	0.431	0.472	0.441	0.428	0.431	0.437	0.425	0.405	0.372	0.361	
	taud	2.333	2.234	2.136	2.042	2.205	2.288	2.294	2.273	2.293	2.325	2.361	2.349	
	Ebn at Noon	776	805	812	804	841	851	844	827	811	782	758	743	
	Edh at Noon	89	113	139	164	142	131	129	128	118	103	86	82	
All-Sky Solar Radiation	RadAvg	1.55	2.28	3.49	4.79	6.02	6.86	7.10	6.21	4.64	3.07	2.07	1.45	
	RadStd	0.22	0.33	0.33	0.55	0.54	0.60	0.27	0.40	0.37	0.28	0.27	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)		+0.66	N/A	N/A	+1.18	N/A	N/A	N/A	N/A	N/A	N/A
		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-90	N/A	N/A

Nomenclature: See separate page